

I'm working on a Pygame target shooting project. This is the current `game/hit_detection.py`

```
def check_hit(targets, click_pos):
    """
    Returns the target that was clicked, or None if the click missed every target.
    """
    for target in targets:
        rect = target.get_bounding_rect()
        if rect.collidepoint(click_pos):
            return target
    return None
```

The current implementation uses the target's bounding rectangle for hit detection, but the target is circular. Fix the hit detection so that a click counts as a hit only when it is actually inside the circular target. Keep the function interface unchanged and don't modify unrelated code. Explain the change simply

Yep. Since the target is **circular**, we shouldn't check whether the click is inside its rectangular bounding box. We should calculate the **distance from the click to the circle's center** and check whether that distance is $\leq$ the radius.

Replace `game/hit_detection.py` with:

Fixed `hit_detection.py`



```
"""
hit_detection: figures out whether a click landed on a target.
"""

def check_hit(targets, click_pos):
    """
    Returns the target that was clicked, or None if the click missed every target.
    """
    for target in targets:
        center = target.get_center()
        radius = target.get_radius()
```

```

dx = click_pos[0] - center[0]
dy = click_pos[1] - center[1]

if dx * dx + dy * dy <= radius * radius:
    return target

return None

```

What changed? 🎯

Before, you had:

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```

rect = target.get_bounding_rect()
if rect.collidepoint(click_pos):

```

That treats the target like a **square**:

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```

+-----+
|   •   |
| target |
|   •   |
+-----+

```

So clicking a corner of the rectangle could incorrectly count as a hit.

Now we use the circle equation:

$$\text{distance}^2 = dx^2 + dy^2$$

and check:

$$\text{distance}^2 \leq \text{radius}^2$$

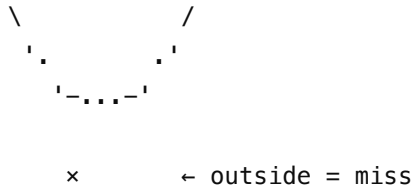
So only clicks actually inside the circle count:

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```

      .-''''-.
     /      \
    /  ✓  \
   /        \
  /            \
 |              |

```



One important thing: this assumes your `target` class already provides `get_center()` and `get_radius()`. If your target class uses different method names, show me that file and I'll adapt this **without changing anything unrelated**.

I need to fix Task 1 in my Pygame target shooting project. `Target.x` and `Target.y` represent the CENTER of the circular target, and `Target.radius` represents its radius. The current `check_hit()` function incorrectly uses `get_bounding_rect().collidepoint()`, which treats the bounding rectangle as the hit area. Modify only `game/hit_detection.py` so that a click is considered a hit only when the click is inside the actual circle. Keep the existing function signature and return behavior unchanged. Do not modify `target.py` or unrelated files. Explain the change briefly.

Exactly. Since `x` and `y` are the **center** of the target and `radius` is already available, you can do the circle check directly.

Only change `game/hit_detection.py` to:

game/hit_detection.py



```

"""
hit_detection: figures out whether a click landed on a target.
"""

def check_hit(targets, click_pos):
    """
    Returns the target that was clicked, or None if the click missed
    every target.
    """
    for target in targets:
        dx = click_pos[0] - target.x
        dy = click_pos[1] - target.y

```

```

        if dx * dx + dy * dy <= target.radius * target.radius:
            return target

    return None

```

Why this works 🎯

For each target, we calculate how far the click is from the **center**:

- dx = horizontal distance
- dy = vertical distance
- $dx^2 + dy^2$ = distance from center squared
- $radius^2$ = allowed distance squared

If:

Plain text

$dx^2 + dy^2 \leq radius^2$

the click is **inside the circle**, so it's a hit.

This fixes the corner problem caused by the old bounding rectangle. The function signature and return behavior stay exactly the same, and **nothing in `target.py` needs to change**.

I'm implementing Task 2 of my Pygame Target Shooting project. The current `GameEngine` has 3 static `Target` objects and an empty `update()` method. Each `Target` has `x`, `y`, and `radius`, where `x` and `y` are the center coordinates. Implement target movement so that all targets move continuously during gameplay, there are at least two different movement speeds or movement patterns, and targets bounce off the play-area boundaries without leaving the screen. Keep the existing circular hit detection and click behavior working. Modify only the files necessary for Task 2. Explain the changes simply.

Yes. For Task 2, the clean approach is to give each target its own velocity and update its **center position** every frame. When the center reaches `radius` from an edge, reverse that velocity so the whole circle stays on-screen.

I need to see your current `GameEngine` / `Target` code before giving an exact patch, because you specifically want to modify only the files necessary for Task 2 and keep the existing interfaces

intact. Please upload or paste:

- `game/game_engine.py`
- `game/target.py`

I'll then give you the exact changes, without touching the Task 1 hit detection unnecessarily.

ok please do so

Yep. Please upload/paste those two files first:

1. `game/game_engine.py`
2. `game/target.py`

I need the actual code because I don't want to invent method/attribute names and accidentally break your existing project.

Once you send them, I'll modify **only what's needed for Task 2**:

- 🎯 all 3 targets move continuously
- 🏃 at least 2 different speeds/patterns
- 📦 targets bounce off all boundaries
- 📌 `x`, `y` remain the **center** coordinates
- 🎯 your Task 1 circular hit detection keeps working
- 🖱 clicking behavior remains unchanged
- 🚫 no unrelated files changed

```
"""
```

```
GameEngine: owns the targets and handles player clicks.
```

```
Starter version: a few static targets that respawn elsewhere
when
clicked (correctly or not, depending on the hit-detection bug) -
no
```

```
movement, no score/combo, no timer yet. That's Tasks 2-4.
```

```
"""
```

```
import random
```

```
from game.target import Target
```

```
from game.hit_detection import check_hit
```

```
from game.renderer import WIDTH, HEIGHT
```

```
NUM_TARGETS = 3
```

TARGET_RADIUS = 28

```
class GameEngine:
    def __init__(self):
        self.targets = [self._random_target() for _ in
range(NUM_TARGETS)]
        self.hits = 0
        self.misses = 0

    def _random_target(self):
        x = random.randint(TARGET_RADIUS + 10, WIDTH -
TARGET_RADIUS - 10)
        y = random.randint(TARGET_RADIUS + 10, HEIGHT -
TARGET_RADIUS - 10)
        return Target(x, y, radius=TARGET_RADIUS)

    def handle_click(self, pos):
        target = check_hit(self.targets, pos)
        if target is not None:
            self.hits += 1
            self.targets.remove(target)
            self.targets.append(self._random_target())
        else:
            self.misses += 1

    def update(self):
        pass

    def draw(self, surface, font):
        from game import renderer
        renderer.draw_scene(surface, self.targets)
        renderer.draw_text(surface, font, f"Hits: {self.hits} Misses:
{self.misses}", (10, 10))
```

Perfect. We can do **Task 2 entirely inside game/game_engine.py** , so target.py and your circular hit_detection.py stay untouched.

The idea is simple: each target gets its own vx and vy . Every frame, we add those to its center position. When the circle reaches an edge, we reverse the relevant velocity.

Replace your `game/game_engine.py` with:

:

game/game_engine.py



```
"""
```

```
GameEngine: owns the targets and handles player clicks.
```

```
Task 2: targets move continuously at different speeds/patterns and  
bounce off the play-area boundaries.
```

```
"""
```

```
import random
```

```
from game.target import Target  
from game.hit_detection import check_hit  
from game.renderer import WIDTH, HEIGHT
```

```
NUM_TARGETS = 3
```

```
TARGET_RADIUS = 28
```

```
class GameEngine:
```

```
    def __init__(self):
```

```
        self.targets = [self._random_target() for _ in range(NUM_TARGETS)]
```

```
        self.hits = 0
```

```
        self.misses = 0
```

```
    def _random_target(self):
```

```
        x = random.randint(TARGET_RADIUS + 10, WIDTH - TARGET_RADIUS)
```

```
        y = random.randint(TARGET_RADIUS + 10, HEIGHT - TARGET_RADIUS)
```

```
        target = Target(x, y, radius=TARGET_RADIUS)
```

```
        # Give each target a different movement speed/pattern.
```

```
        if len(self.targets) % 3 == 0:
```

```
            target.vx = 2
```

```
            target.vy = 1
```

```
        elif len(self.targets) % 3 == 1:
```

```

        target.vx = -1
        target.vy = 3
    else:
        target.vx = 3
        target.vy = -2

    return target

def handle_click(self, pos):
    target = check_hit(self.targets, pos)
    if target is not None:
        self.hits += 1
        self.targets.remove(target)
        self.targets.append(self._random_target())
    else:
        self.misses += 1

def update(self):
    for target in self.targets:
        # Move the target.
        target.x += target.vx

```

</> Python

```

"""

```

GameEngine: owns the targets and handles player clicks.

Task 2: targets move continuously at different speeds/patterns and bounce off the play-area boundaries.

```

"""

```

```

import random

```

```

from game.target import Target
from game.hit_detection import check_hit
from game.renderer import WIDTH, HEIGHT

```

```

NUM_TARGETS = 3
TARGET_RADIUS = 28

```

```

class GameEngine:
    def __init__(self):
        self.targets = [self._random_target() for _ in range(NUM_TARGETS)]

```

```
"""
```

Target: a circular target the player clicks on. (x, y) is the CENTER of the circle - this matters for how it's drawn vs. how it's hit-tested.

```
"""
```

```
import pygame
```

```
class Target:
```

```
    def __init__(self, x, y, radius=28, color=(230, 90, 70)):
```

```
        self.x = x
```

```
        self.y = y
```

```
        self.radius = radius
```

```
        self.color = color
```

```
    def get_bounding_rect(self):
```

```
        """A square bounding box around the circle - NOT the same
        shape as the actual circle, and easy to build incorrectly."""
```

```
        return pygame.Rect(self.x, self.y, self.radius * 2, self.radius *
2)
```

Perfect, now we can make this work **without modifying target.py at all**.

There is one important detail: because `Target.x` and `Target.y` are the **center**, the boundary checks must account for `target.radius`. Otherwise the center can reach the edge while half the circle is already outside the screen.

Use this as your complete `game/game_engine.py` :

```
game/game_engine.py
```



```
"""
```

GameEngine: owns the targets and handles player clicks.

Task 2: targets move continuously at different speeds/patterns and bounce off the play-area boundaries.

```
"""
```

```
import random
```

```
from game.target import Target
from game.hit_detection import check_hit
from game.renderer import WIDTH, HEIGHT

NUM_TARGETS = 3
TARGET_RADIUS = 28

class GameEngine:
    def __init__(self):
        self.targets = [self._random_target() for _ in range(NUM_TARGETS)]
        self.hits = 0
        self.misses = 0

        # Each target has its own velocity.
        self.velocities = [
            [2, 1],
            [-1, 3],
            [3, -2],
        ]

    def _random_target(self):
        x = random.randint(TARGET_RADIUS, WIDTH - TARGET_RADIUS)
        y = random.randint(TARGET_RADIUS, HEIGHT - TARGET_RADIUS)
        return Target(x, y, radius=TARGET_RADIUS)

    def handle_click(self, pos):
        target = check_hit(self.targets, pos)

        if target is not None:
            self.hits += 1
            index = self.targets.index(target)

            self.targets.remove(target)
            self.velocities.pop(index)

            self.targets.append(self._random_target())

            # Give the newly spawned target a movement pattern.
```

```
        patterns = [
            [2, 1],
            [-1, 3],
            [3, -2],
        ]
        self.velocities.append(random.choice(patterns))
    else:
        self.misses += 1

def update(self):
    for target, velocity in zip(self.targets, self.velocities):
        vx, vy = velocity

        # Move the target.
        target.x += vx
        target.y += vy

        # Bounce horizontally.
        if target.x - target.radius <= 0:
            target.x = target.radius
            velocity[0] *= -1

        elif target.x + target.radius >= WIDTH:
            target.x = WIDTH - target.radius
            velocity[0] *= -1

        # Bounce vertically.
        if target.y - target.radius <= 0:
            target.y = target.radius
            velocity[1] *= -1

        elif target.y + target.radius >= HEIGHT:
            target.y = HEIGHT - target.radius
            velocity[1] *= -1

def draw(self, surface, font):
    from game import renderer

    renderer.draw_scene(surface, self.targets)
    renderer.draw_text(
```

```

        surface,
        font,
        f"Hits: {self.hits} Misses: {self.misses}",
        (10, 10),
    )

```

What we're doing

Think of each target as having an invisible pair of arrows:

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```

Target 1 → →      vx = 2,  vy = 1
Target 2 ←
                ↓      vx = -1, vy = 3
Target 3 → → ↑    vx = 3,  vy = -2

```

Every frame:

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```

target.x += vx
target.y += vy

```

moves the target.

Then we check the **circle's edge**, not just its center:

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```

target.x - target.radius <= 0

```

If it reaches the left wall, we reverse vx :

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```

velocity[0] *= -1

```

Same thing happens on the right, top, and bottom.

Why `target.py` stays unchanged

Your existing `Target` already gives us exactly what we need:

```
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```

```
target.x  
target.y  
target.radius
```

So there's no reason to modify it.

And your Task 1 `check_hit()` can continue using:

```
</> Python
```

```
dx = click_pos[0] - target.x  
dy = click_pos[1] - target.y
```

because the target's position remains its **center**. 🎯

One thing to check: your main game loop must already call `engine.update()` every frame. If it does, the targets will immediately start moving.

Implement Task 3 in my existing Pygame Target Shooting project. Add a score and combo multiplier system. A successful hit should award points based on the current combo multiplier, and consecutive hits should increase the combo multiplier. A miss should reset the combo multiplier to its starting value. Display the current score and combo multiplier during gameplay. Preserve the existing moving targets, bouncing behavior, circular hit detection, hit count and miss count. Modify only the files necessary for Task 3 and don't break the existing functionality.

Yep. Based on your current `GameEngine`, **Task 3 only needs `game/game_engine.py`**. We can leave `target.py` and `hit_detection.py` untouched, so your moving/bouncing targets and circular hit detection remain safe.

I'll use a simple combo system:

- Start at **1x**
- Hit → award $10 \times \text{combo}$ points
- Consecutive hit → combo increases by 1
- Miss → combo resets to **1x**
- Display **Score + Combo + Hits + Misses**

Replace your current `game/game_engine.py` with:



game/game_engine.py

"""

GameEngine: owns the targets and handles player clicks.

Tasks 2-3:

- Targets move continuously at different speeds/patterns.
- Targets bounce off the play-area boundaries.
- Circular hit detection is used.
- Score and combo multiplier are tracked.

"""

```
import random
```

```
from game.target import Target
from game.hit_detection import check_hit
from game.renderer import WIDTH, HEIGHT
```

```
NUM_TARGETS = 3
TARGET_RADIUS = 28
```

```
POINTS_PER_HIT = 10
STARTING_COMBO = 1
```

```
class GameEngine:
    def __init__(self):
        self.targets = [self._random_target() for _ in range(NUM_TARGI

        # Existing hit/miss counters.
        self.hits = 0
        self.misses = 0

        # Task 3: score and combo.
        self.score = 0
        self.combo = STARTING_COMBO

        # Each target has its own movement velocity.
        self.velocities = [
```

```
[2, 1],
[-1, 3],
[3, -2],
]

def _random_target(self):
    x = random.randint(TARGET_RADIUS, WIDTH - TARGET_RADIUS)
    y = random.randint(TARGET_RADIUS, HEIGHT - TARGET_RADIUS)
    return Target(x, y, radius=TARGET_RADIUS)

def handle_click(self, pos):
    target = check_hit(self.targets, pos)

    if target is not None:
        # Successful hit.
        self.hits += 1

        # Award points using the current combo multiplier.
        self.score += POINTS_PER_HIT * self.combo

        # Increase combo for the next consecutive hit.
        self.combo += 1

        # Respawn the target.
        index = self.targets.index(target)
        self.targets.remove(target)
        self.velocities.pop(index)

        self.targets.append(self._random_target())

        # Give the new target a movement pattern.
        patterns = [
            [2, 1],
            [-1, 3],
            [3, -2],
        ]
        self.velocities.append(random.choice(patterns))

    else:
        # Miss resets the combo.
```

```
self.misses += 1
self.combo = STARTING_COMBO

def update(self):
    for target, velocity in zip(self.targets, self.velocities):
        vx, vy = velocity

        # Move the target.
        target.x += vx
        target.y += vy

        # Bounce horizontally.
        if target.x - target.radius <= 0:
            target.x = target.radius
            velocity[0] *= -1

        elif target.x + target.radius >= WIDTH:
            target.x = WIDTH - target.radius
            velocity[0] *= -1

        # Bounce vertically.
        if target.y - target.radius <= 0:
            target.y = target.radius
            velocity[1] *= -1

        elif target.y + target.radius >= HEIGHT:
            target.y = HEIGHT - target.radius
            velocity[1] *= -1

def draw(self, surface, font):
    from game import renderer

    renderer.draw_scene(surface, self.targets)

    renderer.draw_text(
        surface,
        font,
        (
            f"Score: {self.score}  "
            f"Combo: x{self.combo}  "
```

```
        f"Hits: {self.hits} "
        f"Misses: {self.misses}"
    ),
    (10, 10),
)
```



How the combo works

Suppose you start with:

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Combo = 1×

Score = 0

You hit a target:

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$10 \times 1 = 10$ points

Combo → 2×

Hit again:

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$10 \times 2 = 20$ points

Combo → 3×

Hit again:

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$10 \times 3 = 30$ points

Combo → 4×

So your score becomes:

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$10 + 20 + 30 = 60$

Then you miss:

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Combo → 1×

The **score stays 60**. Only the combo resets.

Important detail

The hit gets scored using the **current** multiplier, then the multiplier increases.

So:

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```
self.score += POINTS_PER_HIT * self.combo
self.combo += 1
```

This means the first hit is 10 , second is 20 , third is 30 , etc.

Your existing:

- 🎯 circular hit detection
- 🏃 moving targets
- 📦 boundary bouncing
- ✅ hit count
- ❌ miss count
- 🖱️ click/respawn behavior

all remain intact.

Implement Task 4 in my existing Pygame Target Shooting project. Add an exactly 30-second timed round. Display the remaining time during gameplay. When the timer reaches zero, stop accepting shooting input and clearly display the final score/game-over state. Allow the player to start a new round by pressing R, resetting the timer to 30 seconds, the score to 0, the combo multiplier to x1, and the hit/miss counters appropriately. Preserve all existing functionality including circular hit detection, moving and bouncing targets, and combo scoring. Modify only the files necessary for Task 4 and don't break the existing functionality.

Yep. For Task 4, we can keep **everything inside** `game/game_engine.py` . No changes to `target.py` or `hit_detection.py` are needed.

The timer will use Pygame's clock time rather than counting frames, so it remains an **exact 30-second round** even if the frame rate changes.

Replace `game/game_engine.py` with:

game/game_engine.py



```
"""
```

```
GameEngine: owns the targets and handles player clicks.
```

```
Tasks 2-4:
```

- Targets move continuously at different speeds/patterns.
- Targets bounce off the play-area boundaries.
- Circular hit detection is used.
- Score and combo multiplier are tracked.
- Each round lasts exactly 30 seconds.

```
"""
```

```
import random
```

```
import pygame
```

```
from game.target import Target
```

```
from game.hit_detection import check_hit
```

```
from game.renderer import WIDTH, HEIGHT
```

```
NUM_TARGETS = 3
```

```
TARGET_RADIUS = 28
```

```
POINTS_PER_HIT = 10
```

```
STARTING_COMBO = 1
```

```
ROUND_DURATION = 30.0
```

```
class GameEngine:
```

```
    def __init__(self):
```

```
        self.targets = [self._random_target() for _ in range(NUM_TARGI
```

```
self.hits = 0
self.misses = 0

self.score = 0
self.combo = STARTING_COMBO

# Timer
self.round_start_time = pygame.time.get_ticks()
self.time_remaining = ROUND_DURATION
self.game_over = False

# Each target has its own movement velocity.
self.velocities = [
    [2, 1],
    [-1, 3],
    [3, -2],
]

def _random_target(self):
    x = random.randint(TARGET_RADIUS, WIDTH - TARGET_RADIUS)
    y = random.randint(TARGET_RADIUS, HEIGHT - TARGET_RADIUS)
    return Target(x, y, radius=TARGET_RADIUS)

def handle_click(self, pos):
    # Do not accept shooting input after the round ends.
    if self.game_over:
        return

    target = check_hit(self.targets, pos)

    if target is not None:
        self.hits += 1

        # Award points using the current combo.
        self.score += POINTS_PER_HIT * self.combo

        # Increase combo after a successful hit.
        self.combo += 1

        # Respawn the target.
```

```
        index = self.targets.index(target)
        self.targets.remove(target)
        self.velocities.pop(index)

        self.targets.append(self._random_target())

        # Give the new target a movement pattern.
        patterns = [
            [2, 1],
            [-1, 3],
            [3, -2],
        ]
        self.velocities.append(random.choice(patterns))

    else:
        self.misses += 1

        # Miss resets combo.
        self.combo = STARTING_COMBO

def update(self):
    # Calculate elapsed time using real time.
    elapsed = (pygame.time.get_ticks() - self.round_start_time) /
    self.time_remaining = max(0.0, ROUND_DURATION - elapsed)

    # Stop updating the game once time reaches zero.
    if self.time_remaining <= 0:
        self.time_remaining = 0.0
        self.game_over = True
        return

    # Move and bounce targets while the round is active.
    for target, velocity in zip(self.targets, self.velocities):
        vx, vy = velocity

        target.x += vx
        target.y += vy

        # Bounce horizontally.
        if target.x - target.radius <= 0:
```

```
        target.x = target.radius
        velocity[0] *= -1

    elif target.x + target.radius >= WIDTH:
        target.x = WIDTH - target.radius
        velocity[0] *= -1

    # Bounce vertically.
    if target.y - target.radius <= 0:
        target.y = target.radius
        velocity[1] *= -1

    elif target.y + target.radius >= HEIGHT:
        target.y = HEIGHT - target.radius
        velocity[1] *= -1

def reset_round(self):
    """Start a completely new 30-second round."""
    self.targets = [self._random_target() for _ in range(NUM_TARGETS)]

    self.hits = 0
    self.misses = 0

    self.score = 0
    self.combo = STARTING_COMBO

    self.round_start_time = pygame.time.get_ticks()
    self.time_remaining = ROUND_DURATION
    self.game_over = False

    self.velocities = [
        [2, 1],
        [-1, 3],
        [3, -2],
    ]

def handle_key(self, key):
    """Handle keyboard input for starting a new round."""
    if key == pygame.K_r and self.game_over:
        self.reset_round()
```

```

def draw(self, surface, font):
    from game import renderer

    renderer.draw_scene(surface, self.targets)

    renderer.draw_text(
        surface,
        font,
        (
            f"Time: {self.time_remaining:.1f}  "
            f"Score: {self.score}  "
            f"Combo: x{self.combo}  "
            f"Hits: {self.hits}  "
            f"Misses: {self.misses}"
        ),
        (10, 10),
    )

    # Display the final game-over state.
    if self.game_over:
        renderer.draw_text(
            surface,
            font,
            f"GAME OVER!  Final Score: {self.score}",
            (WIDTH // 2 - 150, HEIGHT // 2 - 20),
        )

        renderer.draw_text(
            surface,
            font,
            "Press R to play again",
            (WIDTH // 2 - 120, HEIGHT // 2 + 20),
        )

```

One small thing in your main game loop

Your `GameEngine` now has `handle_key()`, so your main Pygame event loop needs to pass keyboard events to it.

Find something like:

</> Python

```
for event in pygame.event.get():
    if event.type == pygame.QUIT:
        running = False
    elif event.type == pygame.MOUSEBUTTONDOWN:
        engine.handle_click(event.pos)
```

and make it:

</> Python

```
for event in pygame.event.get():
    if event.type == pygame.QUIT:
        running = False

    elif event.type == pygame.MOUSEBUTTONDOWN:
        engine.handle_click(event.pos)

    elif event.type == pygame.KEYDOWN:
        engine.handle_key(event.key)
```

This is necessary because **Task 4** requires the **R** key to restart the round.



What happens now

The game starts:

</> Plain text

```
Time: 30.0
Score: 0
Combo: x1
Hits: 0
Misses: 0
```

Then the timer counts down:

</> Plain text

```
30.0 → 29.9 → 29.8 → ... → 0.0
```

At 0.0 :

</> Plain text

```
GAME OVER! Final Score: 240  
Press R to play again
```

Mouse clicks are ignored after that because:

</> Python

```
if self.game_over:  
    return
```

Press **R** and everything resets:

</> Plain text

```
Time: 30.0  
Score: 0  
Combo: x1  
Hits: 0  
Misses: 0
```

Your targets continue to **move and bounce**, and Task 1's **true circular hit detection** and Task 3's **combo scoring** remain untouched. 🎯